Beyond Spam Detection: Identifying Psychological Manipulation Tactics in Recruitment Scam Messages with Two-Phase DistilBERT

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***Abstract*—Recruitment scams increasingly unfold in private conversations, where fake recruiters pressure job seekers into paying fees or sharing personal data across many messaging platforms. Existing detectors classify job advertisements or messages from a single channel and return only a scam or legitimate label, while the manipulation tactics described in behavioural studies have remained theoretical. This paper presents BSTD, a platform-agnostic system that detects recruitment scam messages and identifies four psychological manipulation tactics: urgency pressure, fear of missing out (FOMO), sunk-cost influence and social proof. A first DistilBERT model separates scam from legitimate messages, and a second model, initialised from the first through intermediate-task transfer, performs multi-label tactic detection. Messages are accepted from a built-in real-time chat, as pasted text from any platform, or as screenshots through optical character recognition. The models were trained on public, synthetic and 718 real recruiter messages and evaluated with a leakage-aware protocol that groups near-duplicates and back-translations, tests on unseen real messages and adds an external smishing dataset. On real recruiter messages, BSTD reached 81.0% accuracy, a scam-class F1-score of 0.86 and a ROC-AUC of 0.90, compared with 69.8%, 0.75 and 0.81 for a TF-IDF baseline, and it detected 90.8% of external scam messages against 60.1% for the baseline. Tactic detection reached a macro F1-score of 0.83.**

Keywords—recruitment fraud, scam message detection, psychological manipulation tactics, DistilBERT, multi-label classification, intermediate-task transfer, optical character recognition, data leakage

I. Introduction

Online recruitment fraud uses fake job offers to obtain money, personal data or unpaid labour from job seekers [1]. Research and commercial tools have focused mainly on the job advertisement, but the harm is usually done later, in private conversations. After the first contact, a fake recruiter moves the candidate to WhatsApp, Telegram, SMS or e-mail and asks for a registration fee, training costs, copies of documents or bank details. Pitumpe and Rahmati collected more than 29,000 recruitment scam messages and found heavy reuse of message templates [2], which suggests that the language of these conversations has patterns a classifier can learn. In a preliminary survey of 78 job seekers in Sri Lanka carried out for this study, 68% reported receiving suspicious messages from someone claiming to be a recruiter, most often on WhatsApp, LinkedIn and Telegram, and the most requested feature was an explanation of why a message is suspicious.

These conversations rely on psychological pressure rather than on suspicious links alone. A scam message may set an artificial deadline, claim that other candidates have already paid, refer to money the victim has already spent, or show fake endorsements. Building on classic persuasion research [3], Anagha et al. [4] modelled these behaviours in job scams as four signals: urgency pressure, fear of missing out (FOMO), sunk-cost influence and social proof. Their study, however, is a behavioural analysis of survey responses; it does not provide a system that recognises these tactics in real messages.

Existing automated detectors leave three gaps. First, recruitment-fraud detectors classify job advertisements [1], [5], not the conversations that follow them. Second, scam detectors for SMS, e-mail and instant messages [6], [7], [8] each target a single channel and return a binary label or a risk score; none tells the user which manipulation tactic is being used. Third, commercial protections such as Google Messages Scam Detection [9] and Truecaller fraud protection [10] are tied to one application and region. A job seeker who receives a suspicious recruiter message on another platform therefore has no tool that explains the manipulation. In addition, scam datasets are usually assembled from several sources and augmented, which makes the evaluation vulnerable to leakage between training and test data [11].

This paper presents BSTD (Behaviour-inspired Semantic Tactic Detection), a platform-agnostic system that detects recruitment scam messages and names the manipulation tactics they use. Its contributions are as follows:

- Tactic-level detection, from theory to implementation: to the best of our knowledge, BSTD is the first automated detector that turns the four behavioural tactics of recruitment scams into a multi-label classifier for recruiter–candidate messages.
- A two-phase DistilBERT architecture in which a scam classifier serves as an intermediate task that initialises the tactic classifier.
- A platform-agnostic design that accepts messages from a built-in real-time chat, as pasted text from any platform, and as screenshots through optical character recognition (OCR), without access to any platform interface.
- A leakage-aware evaluation protocol that groups near-duplicates and back-translations, tests on unseen real recruiter messages, adds an external smishing dataset, and compares against a TF-IDF baseline with significance testing over three random seeds.

Section II reviews related work, Section III describes the data, Section IV presents the method and system, Section V

gives the experimental setup, Section VI reports the results, Section VII discusses the findings and limitations, and Section VIII concludes the paper.

II. Related Work

A. Recruitment Fraud Detection

Vidros et al. [1] characterised online recruitment fraud and released the Employment Scam Aegean Dataset (EMSCAD), which became the standard benchmark for detecting fraudulent job advertisements. Later work applied machine learning and transformer models to the same task; Taneja et al. [5], for example, proposed Fraud-BERT, a context-aware transformer detector. These approaches assess the advertisement, not the conversation that follows. Pitumpe and Rahmati [2] characterised message-based job scams at scale, but did not build a classifier that warns job seekers during a conversation.

B. Scam and Phishing Message Detection

Ghourabi [6] showed that a BERT-based model detects SMS phishing more reliably than keyword filters. Uddin and Sarker [7] fine-tuned a transformer for phishing e-mail detection and explained its decisions with LIME [12], with explanations aimed at analysts. Tan et al. [8] proposed a framework in which a fine-tuned language model anticipates scammer replies to warn users of messaging scams. Shinde et al. [13] extracted SMS text from images with OCR before classification, which showed the value of screenshot input, but they used classifiers without pre-trained transformers. Public corpora such as the UCI SMS Spam Collection [14], the SMS phishing dataset of Mishra and Soni [15] and Smishtank [16] support this line of work. All of these systems produce a binary label or a risk score; none identifies the manipulation tactic.

C. Manipulation Tactics in Scams

Cialdini [3] described persuasion principles such as social proof and scarcity. Anagha et al. [4] operationalised urgency, FOMO, sunk-cost influence and social proof as behavioural signals in job scams. They found that urgency cues were strongly associated with payment, whereas FOMO cues were difficult to identify reliably. Their findings provide the tactic taxonomy used in this work, but not an automated detector.

D. Transformers, Transfer and Evaluation

BERT [17] and its distilled version DistilBERT [18] provide contextual text representations. DistilBERT is reported to be 40% smaller and 60% faster than BERT while retaining 97% of its language understanding [18], which suits real-time classification. Intermediate-task transfer, that is, fine-tuning on a related task before the target task, is most helpful when the two tasks are related [19]. Leakage, in which information about the test data reaches the training process, is a frequent cause of over-optimistic results [11]. Table I compares the closest studies with BSTD.

TABLE I. Comparison with the Closest Related Work

Study	Input	Channels	Tactics	Explanation
Vidros et al. [1]	Job adverts	–	No	No
Taneja et al. [5]	Job adverts	–	No	No
Ghourabi [6]	SMS	Single	No	No
Uddin and Sarker [7]	E-mail	Single	No	Analyst
Shinde et al. [13]	SMS images	Single	No	No
Tan et al. [8]	Instant messages	Single	No	Yes
Anagha et al. [4]	Survey	–	Theory only	–
BSTD (this work)	**Chat, text, screenshots**	**Any**	**Four tactics**	**User-level**

III. Dataset

A. Sources and Labels

The corpus combines six kinds of data. (i) 718 real recruiter messages collected from LinkedIn, Reddit and other online platforms where job seekers shared messages they had received. (ii) Public SMS phishing and spam messages [15], [14]. (iii) Fraudulent and genuine job advertisements from EMSCAD [1], which add recruitment vocabulary. (iv) Synthetic recruiter messages generated from templates to cover each tactic and reviewed before use. (v) Back-translated paraphrases of real messages through Sinhala, Tamil and French [20]. (vi) Smishtank [16], 1,062 user-reported smishing messages that were kept as an external test set and never used for training. Names, telephone numbers, e-mail addresses and identity numbers were removed from the real messages before analysis, in line with the Personal Data Protection Act of Sri Lanka [21].

Every message has a binary label (scam or legitimate). Messages from tactic-annotated sources also have four binary tactic labels, so that one message can carry several tactics. The labels follow the taxonomy of Anagha et al. [4]: urgency pressure creates artificial deadlines or time pressure; FOMO suggests that the opportunity will be lost to others or is limited; sunk-cost influence uses money or effort already spent to demand further payment; and social proof uses fake authority, endorsements or claims about other candidates.

B. Cleaning and Grouping

The raw collection contained 22,730 messages. Template placeholders that appeared only in synthetic data, such as a bracketed candidate name, were replaced with neutral values (3,556 replacements) so that the model could not learn a source artefact instead of scam language. Messages that were too short (114) and exact duplicates (3,121) were removed. Near-duplicates were detected with TF-IDF cosine similarity of at least 0.90, giving 3,935 pairs, and were merged into groups with a union-find structure; each real message was placed in the same group as its back-translations. The cleaned corpus contains 19,495 messages in 15,724 groups.

C. Leakage-Aware Split

All splits are made by group, so that look-alike messages never appear on both sides of a split. The real test set contains only original real recruiter messages, with no back-translations, synthetic or public data: 63 messages (44 scam and 19 legitimate) that the models never saw during training. Public and synthetic groups were divided into training and validation (90/10), and a further group-disjoint 11% of their training groups was held out as an in-distribution test set. Smishtank, with 973 messages after de-duplication, is the external test set. A final check removed every training

message with a cosine similarity of 0.90 or more to any evaluation message. After this step, the highest similarity between a training message and any evaluation message was 0.899, and the highest similarity between a real test message and the training data was 0.667. To prevent leakage, 131 back-translated copies of real validation and test messages were excluded, and the final similarity check removed 3 training messages, giving the 19,361 messages in Table II. Both phases use the same split, so neither model is trained on the other's test messages. Table II gives the sizes of the splits.

TABLE II. DATA SPLITS

Split	Phase 1 (legitimate / scam)	Phase 2
Training	14,734 (9,153 / 5,581)	7,710
Validation	1,874 (1,078 / 796)	999
Real test	63 (19 / 44)	63
Held-out test	1,717 (1,084 / 633)	856
External (Smishtank)	973 (0 / 973)	–

IV. METHOD

A. Phase 1: Scam Classification

A message x, truncated to 128 tokens, is encoded by DistilBERT, and a classification head maps the [CLS] representation h to a scam probability $p = \text{softmax}(Wh + b)_1$. All layers were fine-tuned with a class-weighted cross-entropy loss (weights 0.81 for legitimate and 1.30 for scam messages), and the real training messages were repeated three times so that real language had more influence than synthetic language. A message is classed as a scam when $p \geq \tau$, where the threshold is chosen on the validation set to maximise the F1-score [22]:

$$\tau^* = \arg\max_\tau F1_{val}(\tau), \quad \tau \in \{0.05, 0.06, \ldots, 0.95\} \tag{1}$$

The selected threshold was $\tau^* = 0.26$, below the default of 0.5, which favours catching scams over avoiding false alarms.

B. Phase 2: Tactic Detection with Intermediate-Task Transfer

Tactic detection is a multi-label problem with a target vector $y \in \{0, 1\}^4$. Instead of starting from the general DistilBERT checkpoint, the Phase 2 encoder is initialised from the fine-tuned Phase 1 model, because every tactic is a property of scam language and related intermediate tasks transfer best [19]. A new dropout and linear layer produces four logits z_k, and the model is trained with a weighted binary cross-entropy loss:

$$L = -¼ \Sigma_k [w_k y_k \log \sigma(z_k) + (1 - y_k) \log(1 - \sigma(z_k))] \tag{2}$$

where σ is the sigmoid function and $w_k = \min(N_k^- / N_k^+, 10)$ is the ratio of negative to positive training examples of tactic k, capped at 10. The resulting weights were 4.37 for urgency, 5.84 for FOMO, 6.82 for sunk-cost influence and 5.88 for social proof. Each tactic has its own decision threshold, selected on the validation set to maximise that tactic's F1-score [23].

C. Training Details

Both phases used AdamW [24] with a learning rate of 2×10^{-5}, weight decay of 0.01, a linear schedule with 10% warm-up, dropout of 0.3, gradient clipping at 1.0 and mixed precision. Phase 1 was trained for 3 epochs with a batch size of 32, and Phase 2 for 6 epochs with a batch size of 16. The best epoch was selected by validation macro F1. Each phase was trained with three random seeds (42, 123 and 2024), and the seed with the best validation score was kept as the final model. The implementation uses PyTorch and Hugging Face Transformers [25] on an NVIDIA T4 GPU; one Phase 1 run took about 3 minutes and one Phase 2 run about 4 minutes.

D. Baseline

The baseline uses TF-IDF features of unigrams and bigrams (minimum document frequency of 2, sublinear term frequency) with logistic regression (C = 1, balanced class weights) [26]. For Phase 2, one logistic regression is trained per tactic. Its thresholds were tuned on the validation set in the same way as for DistilBERT.

E. System Design

Fig. 1 shows the deployed system. Both models run in a FastAPI service and are loaded once at start-up. Messages arrive through three channels. In the built-in recruiter–candidate chat, which uses Socket.IO, every recruiter message is classified when it is delivered; a scam result shows a red warning in the conversation and triggers a notification through RabbitMQ. Messages from external platforms can be pasted as text, and screenshots are converted to text with EasyOCR [27], with Tesseract [28] as a fallback. Because the input is text or an image, the system needs no access to the messaging platform, which makes it platform-agnostic.

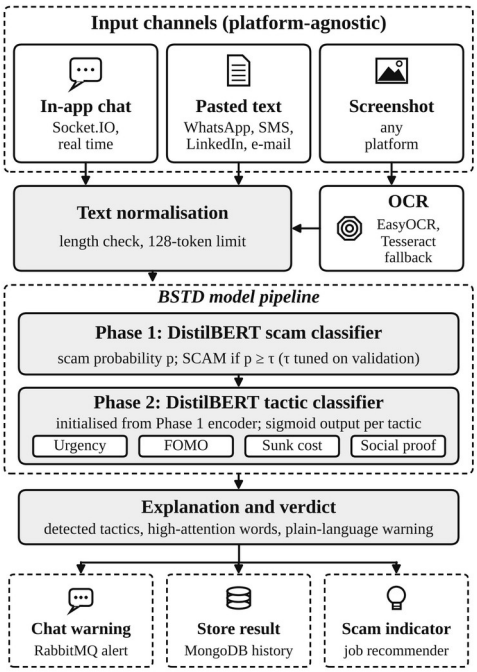


Fig. 1. Architecture of BSTD. Messages from any platform enter through three channels, pass through two DistilBERT models, and return a verdict with tactics and a plain-language warning.

The output contains the verdict, the detected tactics, the five most influential words according to the attention weights of the last encoder layer, and a plain-language warning for each tactic. For urgency pressure, for example, the warning reads: "Legitimate employers never impose same-day deadlines or payment cutoffs." For uncertain scores, the mobile application also shows a "Suspicious" level that asks the user to verify the employer. Each result is stored in MongoDB for the user's history, and a scam indicator is passed to the job recommendation component of the wider platform.

V. EXPERIMENTAL SETUP

The models were evaluated on three test sets. The real test set (63 unseen real recruiter messages) measures real-world

performance and is the main result. The held-out test set (1,717 messages) measures performance on unseen groups from the training sources. The external Smishtank set (973 messages) measures generalisation to user-reported scams from a different source; because it contains only scams, recall is reported for it.

For Phase 1, the metrics are accuracy, precision, recall and F1-score for the scam class, macro F1 and ROC-AUC. For Phase 2, they are per-tactic precision, recall and F1, macro and micro F1, exact-match accuracy and Hamming loss. Confidence intervals of 95% were estimated with 2,000 bootstrap resamples. DistilBERT and the baseline were compared with McNemar's test on paired predictions [29], using the exact binomial form when there were few discordant pairs and the chi-square form otherwise. Results are reported for the final model, together with the mean and standard deviation over the three seeds.

VI. Results

A. Scam Detection

Table III summarises the Phase 1 results. On the real test set, DistilBERT reached an accuracy of 81.0% (95% CI 69.8–90.5%), a scam-class F1-score of 0.864 and a ROC-AUC of 0.901, compared with 69.8%, 0.747 and 0.810 for the baseline. The largest difference is in scam recall: DistilBERT detected 38 of the 44 real scam messages, whereas the baseline missed 16 of them (Fig. 3). The baseline had a slightly higher scam precision (0.903 against 0.864) because it labelled fewer messages as scams; for a protective tool, however, a missed scam is the more costly error. The ROC curves in Fig. 2 show that DistilBERT achieves a higher true positive rate than the baseline at most false positive rates.

TABLE III. Phase 1 Results: Baseline versus DistilBERT

Test set	Model	Acc.	P	R	F1	M-F1	AUC
Real (63)	Baseline	0.698	0.903	0.636	0.747	0.687	0.810
	BSTD	0.810	0.864	0.864	0.864	0.774	0.901
Held-out (1,717)	Baseline	0.955	0.973	0.904	0.937	0.951	0.991
	BSTD	0.966	0.939	0.972	0.955	0.964	0.994
External (973)	Baseline	–	–	0.601	0.751	–	–
	BSTD	–	–	0.908	0.952	–	–

P, R and F1 refer to the scam class; M-F1 is macro F1. The external set contains only scam messages.

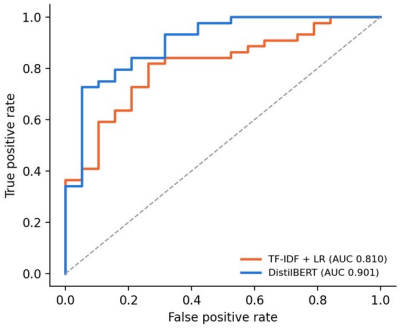


Fig. 2. ROC curves of the baseline and DistilBERT on the real test set.

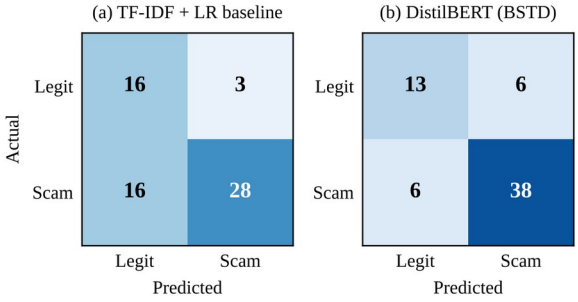


Fig. 3. Confusion matrices on the real test set (63 messages).

The results are stable across seeds: over the three seeds, DistilBERT reached 79.4 ± 2.8% accuracy, a scam F1-score of 0.842 ± 0.032 and a ROC-AUC of 0.897 ± 0.007 on the real test set. On the held-out test set it reached 96.6% accuracy (96.5 ± 0.4% over seeds) and a scam F1-score of 0.955. On the external Smishtank set it detected 90.8% of the 973 scam messages (92.4 ± 1.6% over seeds), compared with 60.1% for the baseline, which shows that the model generalises to user-reported scams from a source it was never trained on.

B. Tactic Detection

TABLE IV. Phase 2 Results on the Held-Out Test Set

Tactic	P	R	F1	Baseline F1	n
Urgency	0.702	0.983	0.819	0.768	180
FOMO	0.721	0.862	0.785	0.804	123
Sunk cost	0.887	0.869	0.878	0.831	99
Social proof	0.830	0.836	0.833	0.820	140
Macro F1			0.829	0.805	
Micro F1			0.824	0.797	
Exact match			0.853	0.815	

P, R and F1 are for BSTD; n is the number of positive messages for each tactic.

Table IV and Fig. 4 report tactic detection on the held-out test set. BSTD reached a macro F1-score of 0.829 (95% CI 0.799–0.856), a micro F1-score of 0.824, an exact-match accuracy of 85.3% and a Hamming loss of 0.061, compared with 0.805, 0.797, 81.5% and 0.067 for the baseline. Over three seeds the macro F1-score was 0.834 ± 0.018. Sunk-cost influence was detected best (F1 0.878). Urgency had a very high recall (0.983) with a lower precision (0.702), so the model rarely misses urgency but sometimes flags it in neutral time references. FOMO was the most difficult tactic (F1 0.785), and the only one for which the baseline was slightly better; this agrees with the behavioural finding that FOMO cues are the hardest to identify reliably [4].

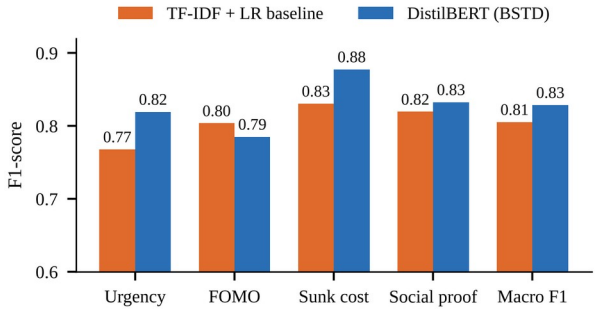


Fig. 4. Per-tactic F1-scores of the baseline and DistilBERT on the held-out test set.

C. Statistical Significance

McNemar's test confirmed that DistilBERT is significantly more accurate than the baseline on the external set (307 against 9 discordant predictions, $p < 0.001$) and on the held-out test set (51 against 32, $p = 0.048$). On the real test set, DistilBERT was correct on 12 messages that the baseline

misclassified, against 5 in the opposite direction; the direction is the same, but with only 63 messages the difference is not significant ($p = 0.14$).

D. Error Analysis

DistilBERT made 12 errors on the real test set. The six missed scams were mostly professionally written internship or interview offers that contain no explicit payment request in the message itself; in such scams the request usually comes later in the conversation. The six false alarms were genuine recruitment messages with urgent or enthusiastic wording, such as short response windows or strong praise of the candidate. A few errors were caused by noisy labels in the collected data, such as a template message with an unfilled placeholder.

VII. Discussion

A. The Value of Leakage-Aware Evaluation

In a preliminary experiment with a random split that did not group near-duplicates, the same architecture reached 93.8% accuracy on a mixed test set. With grouped splits and a test set of real messages only, accuracy on real recruiter messages was 81.0%, while accuracy on held-out data from the training sources remained 96.6%. This gap shows that randomly split or in-distribution results overstate performance on new real conversations, particularly when the data contain templates and back-translations. Scam detection studies should therefore report group-aware splits and a test set of real messages.

B. From Theory to Practice

The tactic classifier shows that the behavioural signals described by Anagha et al. [4] can be detected automatically, with a macro F1-score of 0.83, and the order of difficulty matches their findings. Naming the tactic allows the warning to explain why a message is dangerous, which was the feature most requested in our survey. Because the input is text or a screenshot, the same detector serves messages from WhatsApp, Telegram, LinkedIn, SMS and e-mail without platform integration, unlike channel-bound commercial tools [9], [10].

C. Limitations

The study has several limitations. First, the real test set is small (63 messages), so its confidence intervals are wide and the difference from the baseline on it is not yet statistically significant. Second, the real test set contains few examples of each tactic (5 to 20), so tactic performance was measured mainly on held-out data, and tactic detection on real conversations needs more annotated real messages. Third, the models work on English text; Sinhala and Tamil appear only through back-translation. Fourth, attention weights give only an approximate indication of word importance [30]. Fifth, the external set contains only scams, so only recall could be measured on it, and no formal user study has yet been carried out. Finally, the tactic labels were assigned without a formal inter-annotator agreement study, and the benefit of initialising Phase 2 from Phase 1 was not isolated with an ablation experiment.

VIII. Conclusion

This paper presented BSTD, a platform-agnostic system that detects recruitment scam messages and identifies the psychological manipulation tactics they use. A two-phase DistilBERT model, with tactic detection initialised from scam detection, reached 81.0% accuracy and a scam F1-score of 0.86 on unseen real recruiter messages, detected 90.8% of external scam messages, and identified four tactics with a macro F1-score of 0.83, outperforming a TF-IDF baseline. A leakage-aware protocol showed how grouped splits and real-only test sets give a realistic estimate of performance. Future work will extend the real, tactic-annotated data, including Sinhala and Tamil messages, evaluate multilingual encoders, validate the thresholds of the Suspicious level, add attribution-based explanations such as SHAP [31], and carry out a user study with job seekers.

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